

Lab Experiment 2

Title

Login to the Operating System and Monitoring the Performance

Aim

1. To learn different methods of logging into the Linux operating system.
 2. To monitor system performance using basic Linux commands.
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Requirements

- A computer with Linux installed (Ubuntu recommended).
 - Access to terminal (CLI) and graphical login (GUI).
 - Sudo or root privileges.
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Theory

1. Logging into Linux

Linux provides two primary ways to log into the system:

- **Graphical Login (GUI):**
 - After boot, the system presents a **Login Screen** (Display Manager such as GDM, LightDM).
 - The user selects their username, enters a password, and gets access to the desktop environment (e.g., GNOME, KDE).
 - **Command-Line Login (CLI):**
 - In **multi-user target (text mode)**, the system presents a **login prompt**.
 - The user enters a username and password to access the shell.
 - Example:
 - login: student
 - Password: *****
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2. Monitoring System Performance

Monitoring ensures the system is running efficiently and resources are being used properly.

Common Linux Commands for Performance Monitoring:

Command	Purpose
whoami	Displays the logged-in user
uptime	Shows how long the system has been running, load averages
w	Displays logged-in users and their processes
who	Lists all logged-in users
top	Displays running processes with CPU/memory usage (dynamic view)
htop	(if installed) User-friendly version of top
free -h	Shows memory (RAM) usage
df -h	Displays disk usage of mounted filesystems
du -sh /path	Shows disk usage of a directory
ps aux	Lists all running processes
vmstat	Provides CPU, memory, and I/O statistics
iostat	Shows CPU and disk I/O statistics (requires sysstat package)

Procedure

1. Login to Linux

- Boot the system and log in using **GUI login screen**.
- Logout and switch to **CLI login** (e.g., Ctrl + Alt + F3 on Ubuntu).
- Enter username and password to access the shell.

2. Check User Session

- Run:
->whoami
->who

->W

- Observe the currently logged-in users.

3. Monitor System Uptime and Load

- Run:

->uptime

- Note how long the system has been running and load averages.

4. Check CPU, Memory, and Processes

- Run:

->top

->free -h

->ps aux

- Identify the top processes consuming CPU/memory.

5. Check Disk Usage

- Run:

->df -h

->du -sh /home

- Observe total disk space and usage by directories.

6. Return back to GUI: Ctrl + Alt + F2 on Ubuntu).

Observations

- The system allows login both in GUI and CLI modes.
- The uptime command shows system run time and load averages.
- The top command dynamically updates process and resource usage.
- Disk and memory usage are successfully monitored with df, du, and free.

Result

Students successfully logged into Linux using GUI and CLI. They monitored system performance using commands for CPU, memory, disk, and process management, thereby gaining practical knowledge of system monitoring.